

Module 5

SPI — Protocol + RTL + Basic Testbench

Learning objectives

- SPI mode 0: SCLK idle low; capture on rising edge; change on falling edge; CS_N active low for fram...
- Signals: sclk, mosi, cs_n; clk_div_tick for timing (no MISO in this baseline).
- RTL architecture: spi_master (start, data_in → sclk, mosi, cs_n, done); clk_div (divider → clk_div_...
- Basic TB: Directed transfers (e.g. 0x55, 0xAA); wait for done; no UVM.

Prerequisites

- Modules 1–4 (spec→RTL, UVM, UART baseline, UART UVM).
- Verilator, Make, C++ compiler. No UVM required for spi_baseline.

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

T1["SPI Protocol (Mode 0)"]

T2["RTL Architecture"]

T1 --> T2

T3["Basic Testbench"]

T2 --> T3

Understand the SPI protocol (mode 0, signals), translate it to RTL (SPI master, clk_div), and verify with a **basic** (non-UVM) directed te...

Overview

- Module 5 is the second **protocol** module (after UART):

Syllabus topics

1. SPI Protocol (Mode 0)

- Signals: sclk (serial clock), mosi (master out, slave in), cs_n (active-low chip select). Optional:...
- Mode 0 (CPOL=0, CPHA=0): SCLK idles low; data captured on rising edge; data changed on falling edge.
- Timing: One bit per clk_div_tick; clk_div_tick derived from system clock via divider (e.g. DIVIDER=...

2. RTL Architecture

- `clk_div`: Divides `clk`; outputs `clk_div_tick` one cycle every `DIVIDER` cycles.
- `spi_master`: On start, loads `data_in` into shift register; asserts `cs_n` low; toggles `sclk` on `clk_div_...`

3. Basic Testbench

- Directed test: Reset release; start=1, data_in=0x55; wait for done; repeat for 0xAA; \$finish. Imple...

Hands-on examples

Module 5 self-check

```
$ ./scripts/module5.sh --check
```

Terminal: module5_check

```
[[0;36m===== [[0m
[[0;36mModule 5: Demo commands (SPI baseline) [[0m
[[0;36m===== [[0m
```

From repo root:

```
# 1) Environment sanity:
verilator --version
make --version
```

```
# 2) Run the SPI baseline example (basic TB, no UVM):
cd module5/examples/spi_baseline
make run
```

```
[[1;33m[INFO] [[0m See module5/EXAMPLES.md and module5/examples/spi_baseline/README.md for more.
```

Example 1: SPI baseline

- ``dut/``: ``spi_master.v``, ``clk_div.v``
- ``top_spi_baseline.sv``: Top with `clk_div`, `spi_master`; directed stimulus and checks in initial blocks.
- ``sim_main.cpp``: C++ harness (clock, reset); simulation runs until ``$finish``.

Demo: SPI baseline

```
$ ./scripts/module5.sh --run
```

Terminal: ex01_spi_baseline

Expected: `[PASS] SPI baseline: 0x55 transfer done`, `[PASS] SPI baseline: 0xAA transfer done`, then

Practice & assessment

Exercises

- Run spi_baseline
- Trace one transfer
- Mode 0 timing
- Optional: Waveforms

Assessment checklist

- Can describe SPI mode 0 (idle low, capture on rising, change on falling) and signals sclk, mosi, cs...
- Can explain the role of spi_master and clk_div in the RTL.
- Can run spi_baseline and interpret the directed test result.
- Ready for Module 6: SPI UVM+SV verification.

Summary & next steps

- Understand the SPI protocol (mode 0, signals), translate it to RTL (SPI master, clk_div), and verif...
- Complete module5/CHECKLIST.md
- Run ./scripts/module5.sh --check
- Next: SPI UVM+SV